

NUR ASMAH SABILA BINTI MOHD FAUZI

52224123476

## WEEK 12 : HANDS ON TEST

There are 5 tasks that you have to do in 2 hours. Using your Trial or Azure for Student account to do these tasks is the best option. However, you can also use a sandbox (not the best option). Each of the task answers must be put into an MS Word or PDF file.

Arrange the answers according to the order. The answers must consist of:- IMPORTANT

NOTE: USE CLOUD SHELL IN YOUR AZURE PORTAL!

1. Write the general steps of each process in order. Copy and paste each question before your answer.
2. Paste the screenshot of the steps in the Azure portal and put them in order under each step mentioned above.
3. Show evidence that you are the one who performed the task ( Your screenshot must also include your MS Azure top right account name and the username prompt in your cloud shell terminal) 3. Submit the file to VLE submission folder.

QUESTION 1 Deploy a web application to Azure App Service using Visual Studio. Name your app service as MyApp-YourStudentID. Your web output must display, "Hi! My name is <put your name here>. I am a Cloud Computing Expert." (If you do not have an Azure account use Azure SandBox from this link, then in the Azure portal open the Cloud Shell) (15 Marks)

### Question 1: Deploy a Web App using Visual Studio

**Objective:** Display Hi! My name is <Your Name>. I am a Cloud Computing Expert.

#### Steps:

1. **Open Visual Studio** → Create a new project → Select **ASP.NET Core Web App**.
2. **Set Project Name:** Use any name (e.g., CloudWebApp).
3. **Modify Code:**
  - Open Index.cshtml or HomeController.cs.
  - Replace content with:

html

CopyEdit

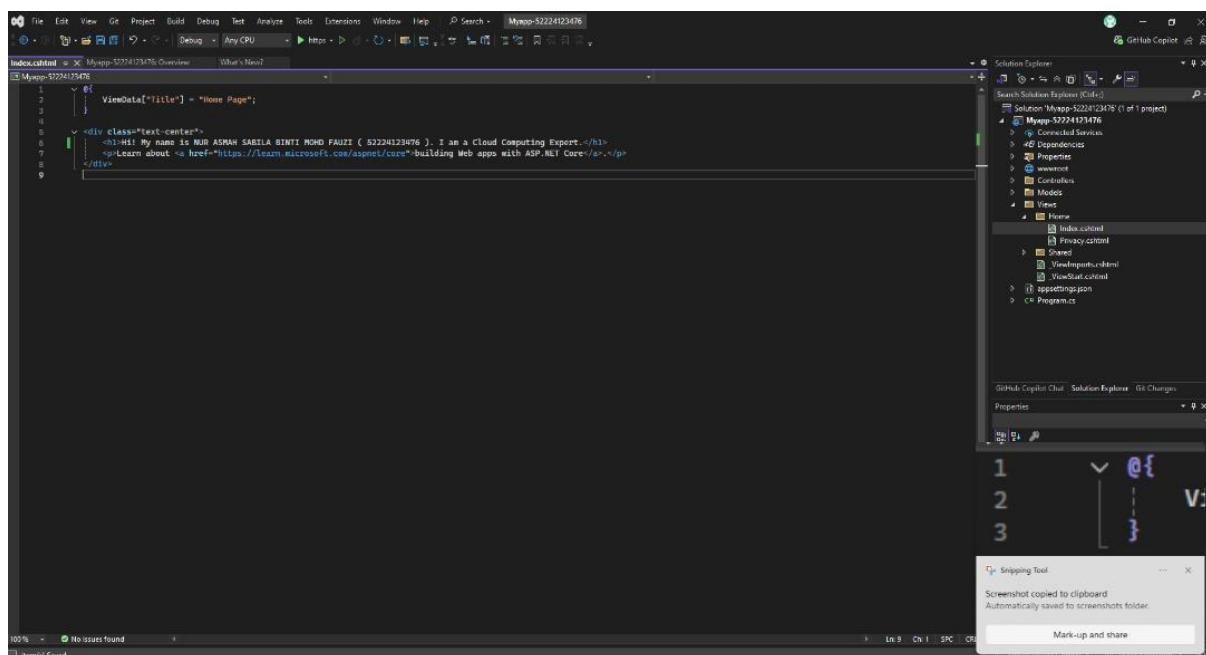
<h1>Hi! My name is NUR ASMAH SABILA BINTI MOHD FAUZI. I am a Cloud Computing Expert.</h1>

#### 4. Publish to Azure:

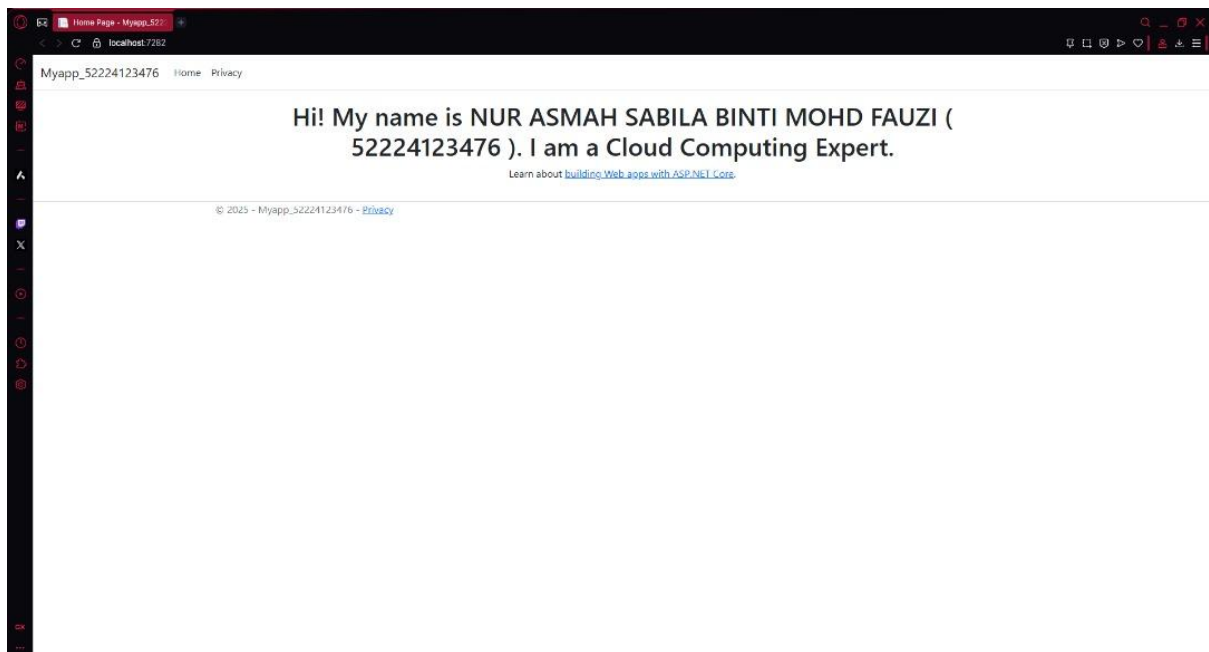
- Right-click the project → Select **Publish**.
- Choose **Azure** → **Azure App Service (Windows)**.
- Click **Create new**.
  - App Name: MyApp-52224123476
  - Resource Group: Create or select existing.
  - Hosting Plan: Select Free (F1).

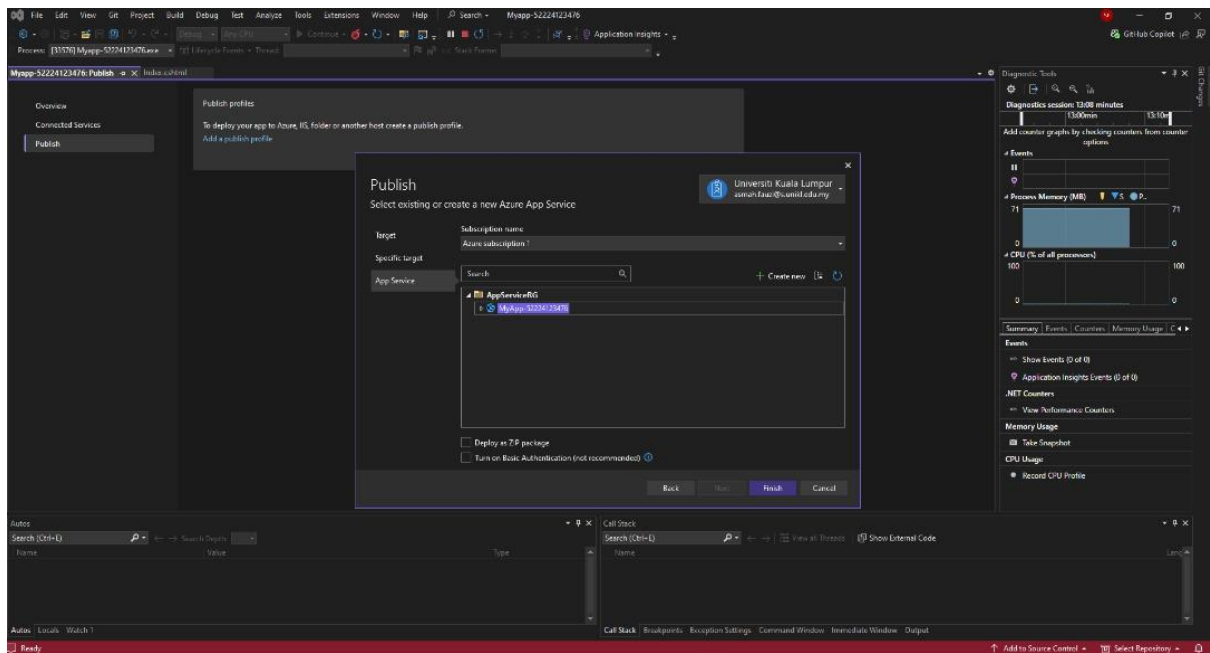
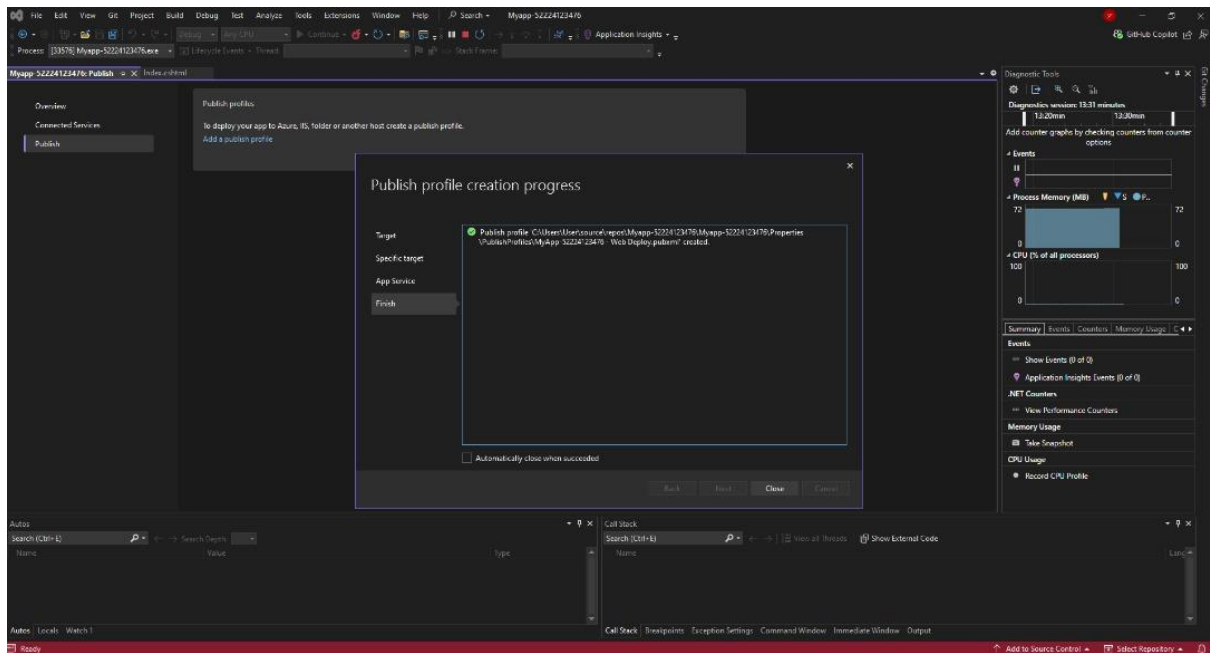
#### 5. Click Publish.

#### 6. Wait for deployment. When done, your app opens in a browser



```
C:\Users\User\source\repos\A x + v - □ x
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: https://localhost:7282
info: Microsoft.Hosting.Lifetime[14]
      Now listening on: http://localhost:5299
info: Microsoft.Hosting.Lifetime[0]
      Application started. Press Ctrl+C to shut down.
info: Microsoft.Hosting.Lifetime[0]
      Hosting environment: Development
info: Microsoft.Hosting.Lifetime[0]
      Content root path: C:\Users\User\source\repos\Myapp-52224123476\Myapp-52224123476
|
```





Visual Studio Code interface showing the Azure App Service extension. The main window displays the "Publish" tab for the project "Myapp-S2224123476". The "Overview" section shows the project name and the "Publish" button. The "Settings" section includes Configuration, Target Framework (net6.0), Deployment Mode (Framework dependent), and Target Runtime (Portable). The "Hosting" section shows the account, subscription, resource group, resource name, and site URL. The "Service Dependencies" section indicates that there are currently no service dependencies configured.

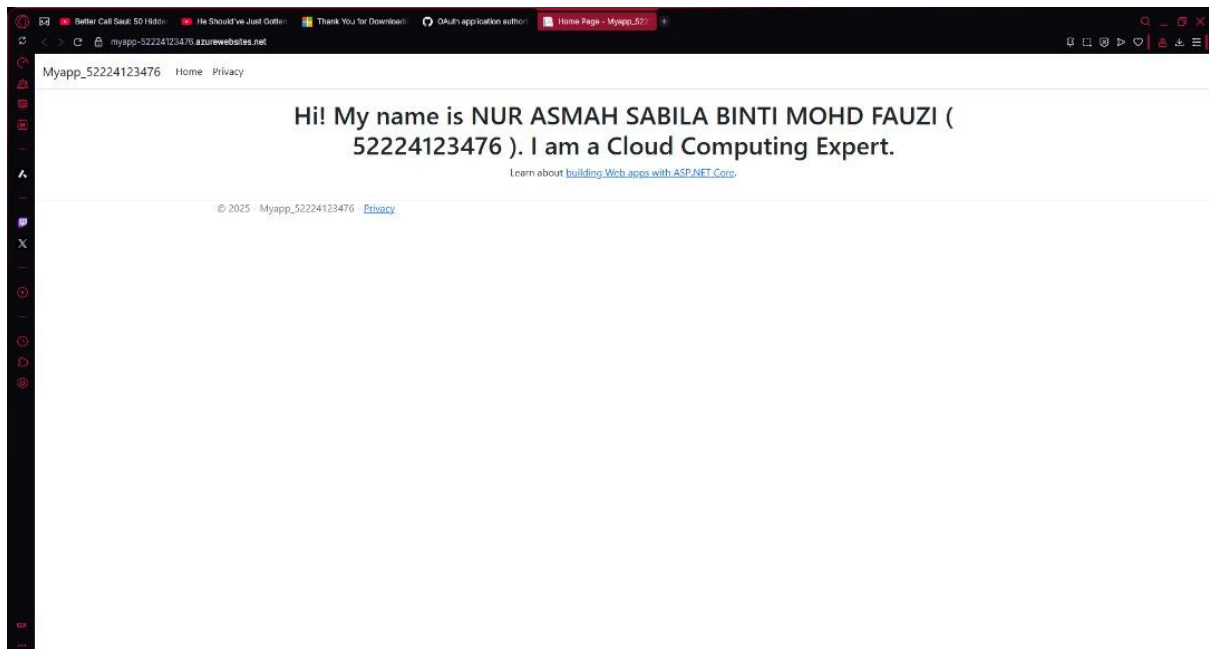
The "Output" window shows the build process output:

```
Build started at 1:05 AM...
1>----- Build started: Project: Myapp-S2224123476, Configuration: Debug Any CPU -----
1>Skipping analyzers to speed up the build. You can choose 'Build' or 'Rebuild' command to run analyzers.
1>Myapp-S2224123476 -> C:\Users\user\source\repos\Myapp-S2224123476\Myapp-S2224123476\bin\Debug\net6.0\Myapp-S2224123476.dll
***** Build: 1 succeeded, 0 failed, 0 up to date, 0 skipped *****
***** Build completed at 1:05 AM and took 09.15 seconds *****
```

Visual Studio Code interface showing the Azure App Service extension. The main window displays the "Publish" tab for the project "Myapp-S2224123476". The "Overview" section shows the project name and the "Publish" button. The "Settings" section includes Configuration, Target Framework (net6.0), Deployment Mode (Framework dependent), and Target Runtime (Portable). The "Hosting" section shows the account, subscription, resource group, resource name, and site URL. The "Service Dependencies" section indicates that there are currently no service dependencies configured.

The "Output" window shows the build process output:

```
Build started at 1:05 AM...
1>----- Build started: Project: Myapp-S2224123476, Configuration: Debug Any CPU -----
1>Skipping analyzers to speed up the build. You can choose 'Build' or 'Rebuild' command to run analyzers.
1>Myapp-S2224123476 -> C:\Users\user\source\repos\Myapp-S2224123476\Myapp-S2224123476\bin\Debug\net6.0\Myapp-S2224123476.dll
***** Build: 1 succeeded, 0 failed, 0 up to date, 0 skipped *****
***** Build completed at 1:05 AM and took 09.15 seconds *****
```



QUESTION 2 Use cloud bash shell to automate the creation of Azure virtual machines. Put VM name as VM-YourStudentID. Show that you can SSH to the VM from the Cloud Shell environment. In Azure Portal show all the VM resources (If you do not have an Azure account use Azure SandBox from this link, then in the Azure portal open the Cloud Shell) (10 Marks)

1. Open **Azure Cloud Shell** (Bash) from the portal.
2. Create a Resource Group:
3. Create a Virtual Machine:
4. Connect to the VM:

```
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rnr [ ~ ]$ az group create --name MyResourceGroupLatest --location westus
{
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest",
  "location": "westus",
  "managedBy": null,
  "name": "MyResourceGroupLatest",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}

rnr [ ~ ]$ az vm create \
--resource-group MyResourceGroupLatest \
--name VM-52224123476 \
--image Ubuntu2204 \
--admin-username azureuser \
--size Standard_B1s \
--location westus \
--generate-ssh-keys
{
  "fqdns": "",
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest/providers/Microsoft.Compute/virtualMachines/VM-52224123476",
  "location": "westus",
  "macAddress": "08-00-3A-50-0B-44",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.184.101.42",
  "resourceGroup": "MyResourceGroupLatest",
  "zones": ""
}

rnr [ ~ ]$
```

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rnr [ ~ ]$ ssh azureuser@172.184.101.42
The authenticity of host '172.184.101.42 (172.184.101.42)' can't be established.
ED25519 key fingerprint is SHA256:bTJGxvthamjKG6VPwskvwpVYefIKZ+2m1hJPY2BTI.
This key is not known by any other names.
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added '172.184.101.42' (ED25519) to the list of known hosts.
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1029-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Tue Jun  3 15:52:12 UTC 2025

System load: 0.07          Processes:            105
Usage of /:  5.3% of 28.89GB Users logged in:      0
Memory usage: 33%          IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

azureuser@VM-52224123476:~$
```

QUESTION 3 Automate the deployment of Azure App Services using Bash. Name your app service as MyApp-YourStudentID. In Azure Portal shows all the VM resources. Browse the App Service in your browser. (If you do not have an Azure account use Azure SandBox from this link, then in the Azure portal open the Cloud Shell) (10 Marks)

### Question 3: Deploy a Web App using Azure CLI (15 Marks)

#### Steps:

1. Define variables:
2. Create Resource Group:

3. Create App Service Plan:
4. Create Web App:
5. Deploy content:
6. Access your app through the link.

```
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Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help  
[ ~ ] $ appName="MyApp-52224123476"  
resourceGroup="AppServiceRG"  
location="westus"  
planName="AppServicePlan52224123476"  
[ ~ ] $ az group create --name $resourceGroup --location $location  
{  
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/AppServiceRG",  
  "location": "westus",  
  "managedBy": null,  
  "name": "AppServiceRG",  
  "properties": {  
    "provisioningState": "Succeeded"  
  },  
  "tags": null,  
  "type": "Microsoft.Resources/resourceGroups"  
}
```

```
Microsoft Azure | Search resources, services, and docs (G+)| Copilot | asmah.fauzi@s.unkl.edu... UNIVERSITI KUALA LUMPUR (UN...  
Switch to PowerShell | Restart | Manage files | New session | Editor | Web preview | Settings | Help  
[ ~ ] $ az appservice plan create --name $planName --resource-group $resourceGroup --sku F1 --location $location  
{  
  "elasticScaleEnabled": false,  
  "extendedLocation": null,  
  "freeOfferExpirationTime": null,  
  "geoRegion": "West US",  
  "hostingEnvironmentProfile": null,  
  "hyperV": false,  
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/AppServiceRG/providers/Microsoft.Web/serverfarms/AppServicePlan52224123476",  
  "isSpot": false,  
  "isXenon": false,  
  "kind": "app",  
  "kubeEnvironmentProfile": null,  
  "location": "westus",  
  "maximumElasticWorkerCount": 1,  
  "maximumNumberOfWorkers": 0,  
  "name": "AppServicePlan52224123476",  
  "numberOfSites": 0,  
  "numberOfWorkers": 0,  
  "perSiteScaling": false,  
  "provisioningState": "Succeeded",  
  "reserved": false,  
  "resourceGroup": "AppServiceRG",  
  "sku": {  
    "capabilities": null,  
    "capacity": 0,  
    "family": "F",  
    "locations": null,  
    "name": "F1",  
    "size": "F1",  
    "skuCapacity": null,  
    "tier": "Free"  
  },  
  "spotExpirationTime": null,  
  "status": "Ready",  
  "subscription": "5bac94b9-510e-4f54-b1e9-6e76ac3a4429",  
  "tags": null,  
  "targetWorkerCount": 0,  
  "targetWorkerSizeId": 0,  
  "type": "Microsoft.Web/serverfarms",  
  "workerTierName": null,  
  "zoneRedundant": false  
}
```

```
tier: "Free"  
},  
"spotExpirationTime": null,  
"status": "Ready",  
"subscription": "5bac94b9-510e-4f54-b1e9-6e76ac3a4429",  
"tags": null,  
"targetWorkerCount": 0,  
"targetWorkerSizeId": 0,  
"type": "Microsoft.Web/serverfarms",  
"workerTierName": null,  
"zoneRedundant": false  
[ ~ ] $
```



```
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Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help
}
nir [ ~ ]$ az webapp create --name $appName --resource-group $resourceGroup --plan $planName
{
  "availabilityState": "Normal",
  "clientAffinityEnabled": true,
  "clientCertEnabled": false,
  "clientCertExclusionPaths": null,
  "clientCertMode": "Required",
  "cloningInfo": null,
  "containerSize": 0,
  "customDomainVerificationId": "6702110E5A9E4E7581713020853799AF4A9245F58303176991EF2F8EAA1A9659",
  "dailyMemoryTimeQuota": 0,
  "deprConfig": null,
  "defaultHostName": "myapp-52224123476.azurewebsites.net",
  "enabled": true,
  "enabledHostNames": [
    "myapp-52224123476.azurewebsites.net",
    "myapp-52224123476.scm.azurewebsites.net"
  ],
  "endToEndEncryptionEnabled": false,
  "extendedLocation": null,
  "ftpPublishingUrl": "Ftps://waws-prod-bay-071.ftp.azurewebsites.windows.net/site/wwwroot",
  "hostNameSslStates": [
    {
      "certificateResourceId": null,
      "hostType": "Standard",
      "ipBasedSslResult": null,
      "ipBasedSslState": "NotConfigured",
      "name": "myapp-52224123476.azurewebsites.net",
      "sslState": "Disabled",
      "thumbprint": null,
      "toUpdate": null,

```

```
    "toUpdateIpBasedSsl": null,
    "virtualIPv6": null,
    "virtualIp": null
  },
  {
    "certificateResourceId": null,
    "hostType": "Repository",
    "ipBasedSslResult": null,
    "ipBasedSslState": "NotConfigured",
    "name": "myapp-52224123476.scm.azurewebsites.net",
    "sslState": "Disabled",
    "thumbprint": null,
    "toUpdate": null,
    "toUpdateIpBasedSsl": null,
    "virtualIPv6": null,
    "virtualIp": null
  }
],
"hostNames": [
  "myapp-52224123476.azurewebsites.net"
],
"hostNamesDisabled": false,
"hostingEnvironmentProfile": null,
"httpsOnly": false,
"hyperV": false,
"id": "/subscriptions/Sbac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/AppServiceRG/providers/Microsoft.Web/sites/MyApp-52224123476",
"identity": null,
"inProgressOperationId": null,
"isDefaultContainer": null,
"isXenon": false,
"keyVaultReferenceIdentity": "SystemAssigned",
"kind": "app",

```

```
"lastModifiedTimeUtc": "2025-06-03T16:01:38.666666",
"location": "West US",
"managedEnvironmentId": null,
"maxNumberOfWorkers": null,
"name": "MyApp-52224123476",
"outboundIpAddresses": "172.184.184.113,23.101.207.250",
"possibleOutboundIpAddresses": "23.101.199.30,23.101.196.70,23.101.201.119,23.101.197.194,172.184.184.113,138.91.194.55,40.78.30.41,40.78.54.188,104.210.43.219,40.78.24.207,20.237.140.37,20.237.142.60,20.237.142.97,20.237.136.27,20.237.137.142,20.237.142.112,168.62.207.25,104.45.223.148,168.61.7.34,168.61.4.187,168.61.1.248,168.62.209.166,23.101.207.250",
"publicNetworkAccess": null,
"redundancyMode": "None",
"repositorySiteName": "MyApp-52224123476",
"reserved": false,
"resourceConfig": null,
"resourceGroup": "AppServiceRG",
"scmSiteAlsoStopped": false,
"serverFarmId": "/subscriptions/Sbac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/AppServiceRG/providers/Microsoft.Web/server-farms/AppServicePlan52224123476",
"siteConfig": {
  "acrUseManagedIdentityCreds": false,
  "acrUserManagedIdentityId": null,
  "alwaysOn": false,
  "antivirusScanEnabled": null,
  "apiDefinition": null,
  "apiManagementConfig": null,
  "appCommandLine": null,
  "appSettings": null,
  "autoHealEnabled": null,
  "autoHealRules": null,
  "autoSwapSlotName": null,
  "azureMonitorLogCategories": null,
  "azureStorageAccounts": null,
  "clusteringEnabled": false,
  "connectionStrings": null,

```

```
"storageType": null,
"supportedTlsCipherSuites": null,
"tracingOptions": null,
"use32BitWorkerProcess": null,
"virtualApplications": null,
"vnetName": null,
"vnetPrivatePortsCount": null,
"vnetRouteAllEnabled": null,
"webSocketsEnabled": null,
"websiteImsZone": null,
"winAuthAdminState": null,
"winAuthTenantState": null,
"windowsConfiguredStacks": null,
"windowsFxVersion": null,
"xManagedServiceIdentityId": null
},
"slotSwapStatus": null,
"state": "Running",
"storageAccountRequired": false,
"suspendedTill": null,
"tags": null,
"targetSwapSlot": null,
"trafficManagerHostNames": null,
"type": "Microsoft.Web/sites",
"usageState": "Normal",
"virtualNetworkSubnetId": null,
"vnetContentShareEnabled": false,
"vnetImagePullEnabled": false,
"vnetRouteAllEnabled": false,
"workloadProfileName": null
}
nur [ ~ ]$
```

```
"cors": null,
"customAppPoolIdentityAdminState": null,
"customAppPoolIdentityTenantState": null,
"defaultDocuments": null,
"detailedErrorLoggingEnabled": null,
"documentRoot": null,
"elasticWebAppScaleLimit": 0,
"experiments": null,
"fileChangeAuditEnabled": null,
"ftpState": null,
"functionAppScaleLimit": null,
"functionsRuntimeScaleMonitoringEnabled": null,
"handlerMappings": null,
"healthCheckPath": null,
"http20Enabled": false,
"http20ProxyFlag": null,
"httpLoggingEnabled": null,
"ipSecurityRestrictions": [
  {
    "action": "Allow",
    "description": "Allow all access",
    "headers": null,
    "ipAddress": "Any",
    "name": "Allow all",
    "priority": 2147483647,
    "subnetMask": null,
    "subnetTrafficTag": null,
    "tag": null,
    "vnetSubnetResourceId": null,
    "vnetTrafficTag": null
  }
],
```



## Your web app is running and waiting for your content

Your web app is live, but we don't have your content yet. If you've already deployed, it could take up to 5 minutes for your content to show up, so come back soon.



 Supporting Node.js, Java, .NET and more

Haven't deployed yet?

Use the deployment center to publish code or set up continuous deployment.

[Deployment center](#)

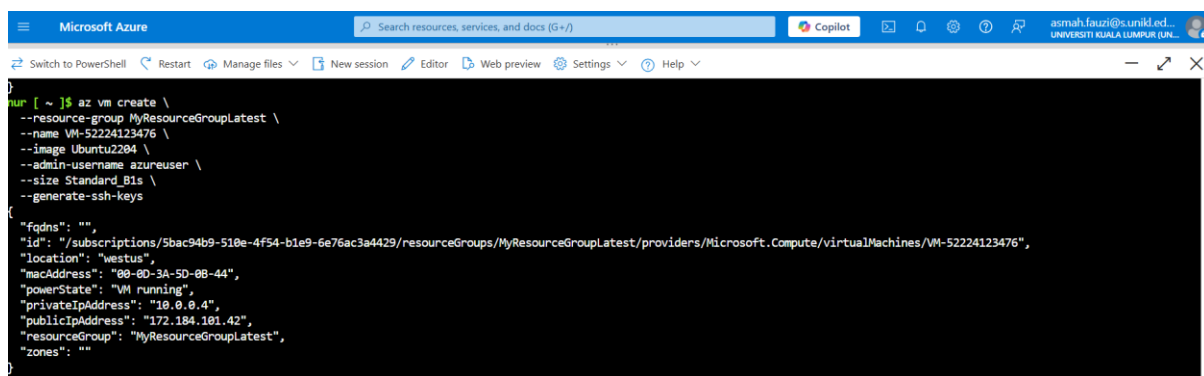
Starting a new web site?

Follow our Quickstart guide to get a web app ready quickly.

[Quickstart](#)

QUESTION 4 Deploy a website using Azure Virtual Machines. Put VM name as VM-YourStudentID. Shows all the VM resources In Azure Portal. Using your browser, the web output must display, "This website was created by <put your name here>". (If you do not have an Azure account use Azure SandBox from this link, then in the Azure portal open the Cloud Shell) (15 Marks)

1. Set variables:
2. Create SQL Database:

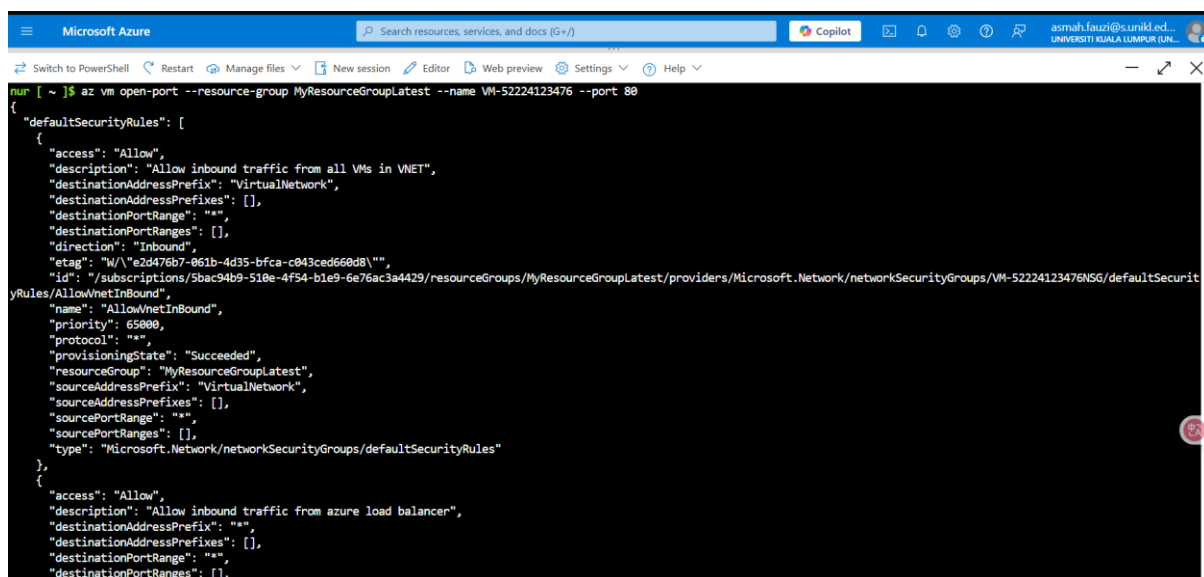


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az vm create \
  --resource-group MyResourceGroupLatest \
  --name VM-52224123476 \
  --image Ubuntu2204 \
  --admin-username azureuser \
  --size Standard_B1s \
  --generate-ssh-keys

{
  "fqdns": "",
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest/providers/Microsoft.Compute/virtualMachines/VM-52224123476",
  "location": "westus",
  "macAddress": "08-0D-3A-5D-08-44",
  "powerState": "VM running",
  "privateIpAddress": "10.0.0.4",
  "publicIpAddress": "172.184.101.42",
  "resourceGroup": "MyResourceGroupLatest",
  "zones": ""
}
```



```
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Switch to PowerShell Restart Manage files New session Editor Web preview Settings Help

az vm open-port --resource-group MyResourceGroupLatest --name VM-52224123476 --port 80

{
  "defaultSecurityRules": [
    {
      "access": "Allow",
      "description": "Allow inbound traffic from all VMs in VNET",
      "destinationAddressPrefix": "VirtualNetwork",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "*",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"e2d476b7-061b-4d35-bfca-c043ced660d8\"",
      "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest/providers/Microsoft.Network/networkSecurityGroups/VM-52224123476/NSG/defaultSecurityRules/AllowVnetInBound",
      "name": "AllowVnetInBound",
      "priority": 65000,
      "protocol": "*",
      "provisioningState": "Succeeded",
      "resourceGroup": "MyResourceGroupLatest",
      "sourceAddressPrefix": "VirtualNetwork",
      "sourceAddressPrefixes": [],
      "sourcePortRange": "*",
      "sourcePortRanges": [],
      "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
    },
    {
      "access": "Allow",
      "description": "Allow inbound traffic from azure load balancer",
      "destinationAddressPrefix": "*",
      "destinationAddressPrefixes": [],
      "destinationPortRange": "*",
      "destinationPortRanges": [],
      "direction": "Inbound",
      "etag": "W/\"e2d476b7-061b-4d35-bfca-c043ced660d8\"",
      "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest/providers/Microsoft.Network/networkSecurityGroups/VM-52224123476/NSG/defaultSecurityRules/AllowLoadBalancerInBound",
      "name": "AllowLoadBalancerInBound",
      "priority": 65001,
      "protocol": "*",
      "provisioningState": "Succeeded",
      "resourceGroup": "MyResourceGroupLatest",
      "sourceAddressPrefix": "AzureLoadBalancer",
      "sourceAddressPrefixes": [],
      "sourcePortRange": "*",
      "sourcePortRanges": [],
      "type": "Microsoft.Network/networkSecurityGroups/defaultSecurityRules"
    }
  ]
}
```

```

    "sourceAddressPrefix": "",
    "sourceAddressPrefixes": [],
    "sourcePortRange": "",
    "sourcePortRanges": [],
    "type": "Microsoft.Network/networkSecurityGroups/securityRules"
  },
  {
    "access": "Allow",
    "destinationAddressPrefix": "",
    "destinationAddressPrefixes": [],
    "destinationPortRange": "80",
    "destinationPortRanges": [],
    "direction": "Inbound",
    "etag": "W/\"e2d476b7-061b-4d35-bfca-c043ced660d8\"",
    "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/MyResourceGroupLatest/providers/Microsoft.Network/networkSecurityGroups/VN-52224123476/NSG/securityRules/open-port-80",
    "name": "open-port-80",
    "priority": 900,
    "protocol": "*",
    "provisioningState": "Succeeded",
    "resourceGroup": "MyResourceGroupLatest",
    "sourceAddressPrefix": "",
    "sourceAddressPrefixes": [],
    "sourcePortRange": "",
    "sourcePortRanges": [],
    "type": "Microsoft.Network/networkSecurityGroups/securityRules"
  }
],
"tags": {},
"type": "Microsoft.Network/networkSecurityGroups"
}
nur [ ~ ]$

```

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```

nur [ ~ ]$ ssh azureuser@172.184.101.42
Welcome to Ubuntu 22.04.5 LTS (GNU/Linux 6.8.0-1029-azure x86_64)

 * Documentation:  https://help.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

System information as of Tue Jun  3 16:17:52 UTC 2025

System load:  0.0      Processes:      106
Usage of /:   5.5% of 28.89GB   Users logged in:  0
Memory usage: 32%      IPv4 address for eth0: 10.0.0.4
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update
New release '24.04.2 LTS' available.
Run 'do-release-upgrade' to upgrade to it.

Last login: Tue Jun  3 15:52:14 2025 from 52.253.87.141
To run a command as administrator (user "root"), use "sudo <command>".
See "man sudo_root" for details.

```

```

azureuser@VN-52224123476:~$ sudo apt update
Hit:1 http://azure.archive.ubuntu.com/ubuntu jammy InRelease
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates InRelease [128 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-backports InRelease [127 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-security InRelease [129 kB]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 Packages [14.1 MB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy/universe Translation-en [5652 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy/universe amd64 c-n-f Metadata [286 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 Packages [217 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse Translation-en [112 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/multiverse amd64 c-n-f Metadata [8372 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 Packages [2619 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main Translation-en [423 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted amd64 Packages [3631 kB]
Get:14 http://azure.archive.ubuntu.com/ubuntu jammy-updates/restricted Translation-en [648 kB]
Get:15 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 Packages [1208 kB]
Get:16 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe Translation-en [298 kB]
Get:17 http://azure.archive.ubuntu.com/ubuntu jammy-updates/universe amd64 c-n-f Metadata [28.7 kB]
Get:18 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 Packages [46.5 kB]
Get:19 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse Translation-en [11.8 kB]
Get:20 http://azure.archive.ubuntu.com/ubuntu jammy-updates/multiverse amd64 c-n-f Metadata [592 B]
Get:21 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 Packages [68.8 kB]
Get:22 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main Translation-en [11.4 kB]
Get:23 http://azure.archive.ubuntu.com/ubuntu jammy-backports/main amd64 c-n-f Metadata [392 B]
Get:24 http://azure.archive.ubuntu.com/ubuntu jammy-backports/restricted amd64 c-n-f Metadata [116 B]
Get:25 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 Packages [30.0 kB]
Get:26 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe Translation-en [16.5 kB]
Get:27 http://azure.archive.ubuntu.com/ubuntu jammy-backports/universe amd64 c-n-f Metadata [672 B]
Get:28 http://azure.archive.ubuntu.com/ubuntu jammy-backports/multiverse amd64 c-n-f Metadata [116 B]
Get:29 http://azure.archive.ubuntu.com/ubuntu jammy-security/main amd64 Packages [2375 kB]
Get:30 http://azure.archive.ubuntu.com/ubuntu jammy-security/main Translation-en [359 kB]
Get:31 http://azure.archive.ubuntu.com/ubuntu jammy-security/restricted amd64 Packages [3513 kB]

```

```
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
26 packages can be upgraded. Run 'apt list --upgradable' to see them.
azureuser@VM-52224123476:~$ sudo apt install -y apache2
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
The following additional packages will be installed:
  apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support ssl-cert
Suggested packages:
  apache2-doc apache2-suexec-pristine | apache2-suexec-custom www-browser bzip2-doc
The following NEW packages will be installed:
  apache2 apache2-bin apache2-data apache2-utils bzip2 libapr1 libaprutil1
  libaprutil1-dbd-sqlite3 libaprutil1-ldap liblua5.3-0 mailcap mime-support ssl-cert
0 upgraded, 13 newly installed, 0 to remove and 26 not upgraded.
Need to get 2142 kB of archives.
After this operation, 8528 kB of additional disk space will be used.
Get:1 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libapr1 amd64 1.7.0-8ubuntu0.22.04.2 [100 kB]
Get:2 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1 amd64 1.6.1-5ubuntu4.22.04.2 [82.8 kB]
Get:3 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-dbd-sqlite3 amd64 1.6.1-5ubuntu4.22.04.2 [11.3 kB]
Get:4 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 libaprutil1-ldap amd64 1.6.1-5ubuntu4.22.04.2 [9170 B]
Get:5 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 liblua5.3-0 amd64 5.3.6-1build1 [140 kB]
Get:6 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-bin amd64 2.4.52-1ubuntu4.14 [1349 kB]
Get:7 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-data all 2.4.52-1ubuntu4.14 [165 kB]
Get:8 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2-utils amd64 2.4.52-1ubuntu4.14 [89.0 kB]
Get:9 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mailcap all 3.70+nmu1ubuntu1 [23.8 kB]
Get:10 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 mime-support all 3.66 [3696 B]
Get:11 http://azure.archive.ubuntu.com/ubuntu jammy-updates/main amd64 apache2 amd64 2.4.52-1ubuntu4.14 [97.9 kB]
Get:12 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 bzip2 amd64 1.0.8-5build1 [34.8 kB]
Get:13 http://azure.archive.ubuntu.com/ubuntu jammy/main amd64 ssl-cert all 1.1.2 [17.4 kB]
```

```
Enabling module env.
Enabling module mime.
Enabling module negotiation.
Enabling module setenvif.
Enabling module filter.
Enabling module deflate.
Enabling module status.
Enabling module reqtimeout.
Enabling conf charset.
Enabling conf localized-error-pages.
Enabling conf other-vhosts-access-log.
Enabling conf security.
Enabling conf serve-cgi-bin.
Enabling site 000-default.
Created symlink /etc/systemd/system/multi-user.target.wants/apache2.service → /lib/systemd/system/apache2.service.
Created symlink /etc/systemd/system/multi-user.target.wants/apache-htcacheclean.service → /lib/systemd/system/apache-htcacheclean.service.
Processing triggers for ufw (0.36.1-4ubuntu0.1) ...
Processing triggers for man-db (2.10.2-1) ...
Processing triggers for libc-bin (2.35-0ubuntu3.9) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

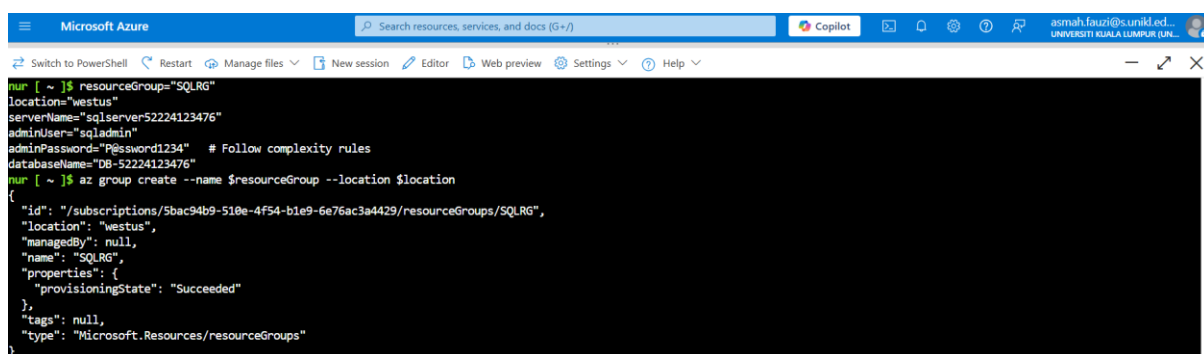
No VM guests are running outdated hypervisor (qemu) binaries on this host.
azureuser@VM-52224123476:~$
```

```
azureuser@VM-52224123476:~$ echo "This website was created by NUR ASMAH SABILA BINTI MOHD FAUZI" | sudo tee /var/www/html/index.html
This website was created by NUR ASMAH SABILA BINTI MOHD FAUZI
azureuser@VM-52224123476:~$ sudo systemctl restart apache2
azureuser@VM-52224123476:~$ exit
```



This website was created by NUR ASMAH SABILA BINTI MOHD FAUZI

QUESTION 5 Deploy an Azure SQL Database using cloud Shell Bash. Put the database name DB-YourStudentID. In Azure Portal Show the database deployed (If you do not have an Azure account use Azure SandBox from this link, then in the Azure portal open the Cloud Shell). (10 Marks



The screenshot shows the Microsoft Azure Cloud Shell interface. The top bar includes the Microsoft Azure logo, a search bar, and a Copilot button. The user's email address, asmah.fauzi@unilid.ed..., is visible in the top right corner. The main area displays a terminal window with the following commands and output:

```
nur [ ~ ]$ resourceGroup="SQLRG"
location="westus"
serverName="sqlserver52224123476"
adminUser="sqladmin"
adminPassword="P@ssword1234" # Follow complexity rules
databaseName="DB-52224123476"
nur [ ~ ]$ az group create --name $resourceGroup --location $location
{
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/SQLRG",
  "location": "westus",
  "managedBy": null,
  "name": "SQLRG",
  "properties": {
    "provisioningState": "Succeeded"
  },
  "tags": null,
  "type": "Microsoft.Resources/resourceGroups"
}
```

```
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PS C:\> az sql db create \
  --name DB-52224123476 \
  --resource-group $ResourceGroup \
  --server $serverName \
  --service-objective S0
{
  "autoPauseDelay": null,
  "availabilityZone": "NoPreference",
  "catalogCollation": "SQL_Latin1_General_CP1_CI_AS",
  "collation": "SQL_Latin1_General_CP1_CI_AS",
  "createMode": null,
  "creationDate": "2025-06-03T16:45:04.863000+00:00",
  "currentBackupStorageRedundancy": "Geo",
  "currentServiceObjectiveName": "S0",
  "currentSku": {
    "capacity": 10,
    "family": null,
    "name": "Standard",
    "size": null,
    "tier": "Standard"
  },
  "databaseId": "df12cc6f-dec8-4205-a5b4-0469ef1bad55",
  "defaultSecondaryLocation": "eastus",
  "earliestRestoreDate": null,
  "edition": "Standard",
  "elasticPoolId": null,
  "elasticPoolName": null,
  "encryptionProtector": null,
  "encryptionProtectorAutoRotation": null,
  "failoverGroupId": null,
  "federatedClientId": null,
  "freelimitExhaustionBehavior": null,

```

```
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PS C:\> az sql server create \
  --name $serverName \
  --resource-group $ResourceGroup \
  --location $location \
  --admin-user $adminUser \
  --admin-password $adminPassword
{
  "administratorLogin": "sqladmin",
  "administratorLoginPassword": null,
  "administrators": null,
  "externalGovernanceStatus": "Disabled",
  "federatedClientId": null,
  "fullyQualifiedDomainName": "sqlserver52224123476.database.windows.net",
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/SQLRG/providers/Microsoft.Sql/servers/sqlserver52224123476",
  "identity": null,
  "isIPv6Enabled": null,
  "keyId": null,
  "kind": "v12.0",
  "location": "westus",
  "minimalTlsVersion": "1.2",
  "name": "sqlserver52224123476",
  "primaryUserAssignedIdentityId": null,
  "privateEndpointConnections": [],
  "publicNetworkAccess": "Enabled",
  "resourceGroup": "SQLRG",
  "restrictOutboundNetworkAccess": "Disabled",
  "state": "Ready",
  "tags": null,
  "type": "Microsoft.Sql/servers",
  "version": "12.0",
  "workspaceFeature": null
}
```

```
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PS C:\> az sql db show \
  --name DB-52224123476 \
  --resource-group $ResourceGroup \
  --server $serverName
{
  "highAvailabilityReplicaCount": null,
  "id": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/resourceGroups/SQLRG/providers/Microsoft.Sql/servers/sqlserver52224123476/databases/DB-52224123476",
  "identity": null,
  "isInfraEncryptionEnabled": false,
  "keys": null,
  "kind": "v12.0,user",
  "ledgerOn": false,
  "licenseType": null,
  "location": "westus",
  "longTermRetentionBackupResourceId": null,
  "maintenanceConfigurationId": "/subscriptions/5bac94b9-510e-4f54-b1e9-6e76ac3a4429/providers/Microsoft.Maintenance/publicMaintenanceConfigurations/SQL_Default",
  "managedBy": null,
  "manualCutover": null,
  "maxLogSizeBytes": null,
  "maxSizeBytes": 268435456000,
  "minCapacity": null,
  "name": "DB-52224123476",
  "pausedDate": null,
  "performCutover": null,
  "preferredEnclaveType": null,
  "readScale": "Disabled",
  "recoverableDatabaseId": null,
  "recoveryServicesRecoveryPointId": null,
  "requestedBackupStorageRedundancy": "Geo",
  "requestedServiceObjectiveName": "S0",
  "resourceGroup": "SQLRG",
  "restoreableDroppedDatabaseId": null,
  "restorePointInTime": null,
  "resumeDate": null,
  "sampleName": null,
  "secondaryType": null,
  "sku": {

```



```
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{
  "performCutover": null,
  "preferredEnclaveType": null,
  "readScale": "Disabled",
  "recoverableDatabaseId": null,
  "recoveryServicesRecoveryPointId": null,
  "requestedBackupStorageRedundancy": "Geo",
  "requestedServiceObjectiveName": "S0",
  "resourceGroup": "SQLRG",
  "restoreableDroppedDatabaseId": null,
  "restorePointInTime": null,
  "resumeDate": null,
  "sampleName": null,
  "secondaryType": null,
  "sku": {
    "capacity": 10,
    "family": null,
    "name": "Standard",
    "size": null,
    "tier": "Standard"
  },
  "sourceDatabaseDeleteDate": null,
  "sourceDatabaseId": null,
  "sourceResourceId": null,
  "status": "Online",
  "tags": null,
  "type": "Microsoft.Sql/servers/databases",
  "useFreeLimit": null,
  "zoneRedundant": false
}
```

Microsoft Azure

Home > Resource groups >

### Resource groups

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+ Create Manage view ...

**You are viewing a new version of Browse experience. Some features may be missing. Click here to access the old experience.**

| Name ↑                    |
|---------------------------|
| AppServiceRG ...          |
| MyResourceGroup ...       |
| MyResourceGroupLatest ... |
| NetworkWatcherRG ...      |
| <b>SQLRG</b> ...          |

Showing 1 - 5 of 5. Display count: 10

#### SQLRG

Resource group

Search

+ Create Manage view Delete resource group Refresh Export to CSV Open query Assign tags ...

**Overview**

- Activity log
- Access control (IAM)
- Tags
- Resource visualizer
- Events
- Settings
  - Deployments
  - Security
  - Deployment stacks
  - Policies
  - Properties
  - Locks
- Cost Management
- Monitoring
- Automation

Add or remove favorites by pressing Ctrl+Shift+F

#### Essentials

JSON View

Resources Recommendations

Filter for any field... Type equals all Location equals all Add filter

Showing 1 to 2 of 2 records. Show hidden types No grouping List view

| Name ↑                                               | Type ↑       | Location ↑ |
|------------------------------------------------------|--------------|------------|
| DB-52224123476 (sqlserver52224123476/DB-52224123476) | SQL database | West US    |
| sqlserver52224123476                                 | SQL server   | West US    |

< Previous Page 1 of 1 Next >

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